

1. Applying Hough Transform for Line Detection

A road lane detection system uses a camera to identify straight lane markings. After edge detection, one edge point is found at coordinate (4, 3). The system applies Hough Transform line detection with angle $\theta = 0^\circ$. This value is used for voting in the accumulator space. Apply the Hough Transform concept and calculate the value of ρ .

- A. 3
- B. 4
- C. 5
- D. 7

Correct Answer: B. 4

Explanation:

$$\rho = x \cos \theta + y \sin \theta$$

Given:

$$\begin{aligned}x &= 4, y = 3, \theta = 0^\circ \\ \cos 0^\circ &= 1, \sin 0^\circ = 0 \\ \rho &= 4(1) + 3(0) = 4\end{aligned}$$

So, the value of ρ is 4.

2. Applying Hough Transform in Document Line Detection

A document image processing system is used to detect table borders in scanned forms. One edge point in the document image is located at (6, 8). The system applies Hough Transform line detection with $\theta = 90^\circ$. The calculated value helps identify possible vertical or horizontal line structures. Use the Hough Transform rule and determine the value of ρ .

- A. 6
- B. 8
- C. 10
- D. 14

Correct Answer: B. 8

Explanation:

$$\rho = x \cos \theta + y \sin \theta$$

Given:

$$\begin{aligned}x &= 6, y = 8, \theta = 90^\circ \\ \cos 90^\circ &= 0, \sin 90^\circ = 1 \\ \rho &= 6(0) + 8(1) = 8\end{aligned}$$

So, the value of ρ is 8.

3. Applying Circle Detection Radius Calculation

An industrial inspection system checks whether bottle caps are circular. A detected circular cap has center point (10, 15), and one boundary point is located at (16, 23). The system calculates the radius to verify whether the object matches the expected cap size. Apply the distance concept and calculate the radius of the detected circle.

- A. 8 pixels
- B. 9 pixels
- C. 10 pixels
- D. 12 pixels

Correct Answer: C. 10 pixels

Explanation:

$$\begin{aligned}r &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\r &= \sqrt{(16 - 10)^2 + (23 - 15)^2} \\r &= \sqrt{6^2 + 8^2} \\r &= \sqrt{36 + 64} = 10\end{aligned}$$

So, the radius is 10 pixels.

4. Applying Accumulator Voting in Hough Transform

A traffic sign detection system uses Hough Transform to detect circular road signs. After the edge points vote in the accumulator space, three possible circle candidates receive vote counts of 45, 30, and 60. The system selects the candidate with the highest vote as the strongest detected circle. Apply the accumulator voting rule and determine the selected candidate.

- A. Candidate with 30 votes
- B. Candidate with 45 votes
- C. Candidate with 60 votes
- D. All candidates are rejected

Correct Answer: C. Candidate with 60 votes

Explanation:

In Hough Transform, the shape candidate with the highest accumulator vote is selected as the strongest detection.

The highest vote count is:

60

So, the selected candidate is Candidate with 60 votes.

5. Applying Object Recognition Accuracy

A factory uses an object recognition system to classify products as bottles, boxes, or packets. During testing, the system checks 250 objects. Out of these, 225 objects are correctly recognized. The engineer calculates accuracy before using the system for product sorting. Apply the accuracy calculation and determine the object recognition accuracy.

- A. 85%
- B. 88%
- C. 90%
- D. 95%

Correct Answer: C. 90%

Explanation:

$$\text{Accuracy} = \frac{\text{Correct Recognitions}}{\text{Total Objects}} \times 100$$
$$\text{Accuracy} = \frac{225}{250} \times 100 = 90\%$$

So, the object recognition accuracy is 90%.

2. Applying Shape Matching Score

A traffic sign recognition system compares a detected road sign with a stored template. The system compares four boundary features, and the absolute differences are 3, 5, 2, and 6. A lower total difference means a better match between the detected sign and the stored template. The system calculates the total matching error before classification. Use the given feature differences and determine the total shape matching error.

- A. 12
- B. 14
- C. 16
- D. 18

Correct Answer: C. 16

Explanation:

$$\text{Total Error} = 3 + 5 + 2 + 6$$
$$\text{Total Error} = 16$$

So, the total shape matching error is 16.

3. Applying Nearest Shape Selection

A robot vision system compares an unknown object shape with three stored shape templates. The matching error with Template A is 18, with Template B is 9, and with Template C is 14. The system selects the template with the lowest matching error as the best match. This helps the robot identify the object category. Apply the shape matching rule and determine the best matching template.

- A. Template A
- B. Template B
- C. Template C
- D. All templates are equal

Correct Answer: B. Template B

Explanation:

In shape matching, the template with the lowest matching error is treated as the best match.

$$\text{Template A} = 18, \text{Template B} = 9, \text{Template C} = 14$$

The lowest error is 9, so the best match is Template B.

4. Applying PCA Mean Calculation

A face recognition system applies Principal Component Analysis to reduce feature dimensions. One simple feature is measured from four face images with values 10, 20, 30, and 40. Before applying PCA, the system calculates the mean value and centers the data. This is an important preprocessing step in PCA. Apply the PCA preprocessing concept and calculate the mean value.

- A. 20
- B. 25
- C. 30
- D. 35

Correct Answer: B. 25

Explanation:

$$\begin{aligned} \text{Mean} &= \frac{10 + 20 + 30 + 40}{4} \\ \text{Mean} &= \frac{100}{4} = 25 \end{aligned}$$

So, the mean value is 25.

5. Applying Dimensionality Reduction Using PCA

An object recognition system extracts 100 features from each image. To reduce computation, PCA is applied and only 20 principal components are retained. The reduced features are then used for classification. The engineer wants to calculate how many features were removed by PCA. Apply the dimensionality reduction concept and determine the number of removed features.

- A. 20
- B. 40
- C. 60
- D. 80

Correct Answer: D. 80

Explanation:

$$\begin{aligned} \text{Removed Features} &= \text{Original Features} - \text{Retained Components} \\ \text{Removed Features} &= 100 - 20 = 80 \end{aligned}$$

So, PCA removes 80 features.

1. Applying Shape Prior Matching Score

A medical image analysis system is used to detect the shape of a kidney from ultrasound images. The system has a stored normal kidney shape model as a shape prior. For one patient image, the system compares five shape features with the stored prior and obtains absolute differences of 4, 3, 5, 2, and 6. A lower total difference means the detected shape is closer to the stored prior. Apply the shape prior matching concept and calculate the total shape difference.

- A. 15
- B. 18
- C. 20
- D. 24

Correct Answer: C. 20

Explanation:

$$\begin{aligned} \text{Total Difference} &= 4 + 3 + 5 + 2 + 6 \\ \text{Total Difference} &= 20 \end{aligned}$$

So, the total shape difference is 20.

2. Applying Best Shape Prior Selection

A traffic sign recognition system uses shape priors to classify road signs. The detected sign is compared with three stored shape priors. The matching error with the circular prior is 12, the triangular prior is 8, and the rectangular prior is 15. The system selects the shape prior with the lowest error as the best match. Use the shape prior matching rule and determine the best matching shape prior.

- A. Circular prior
- B. Triangular prior
- C. Rectangular prior
- D. All priors are equal

Correct Answer: B. Triangular prior

Explanation:

The best matching shape prior is selected based on the lowest matching error.

$$Circular = 12, Triangular = 8, Rectangular = 15$$

The lowest error is 8, so the best matching prior is the triangular prior.

3. Applying CNN Feature Map Size

A deep learning model processes a grayscale image of size 64×64 . A convolution layer uses a 3×3 filter with stride 1 and no padding. The output feature map size is calculated before sending it to the next layer. This helps the developer understand how image dimensions change inside the CNN. Apply the convolution output size concept and determine the output feature map size.

- A. 60×60
- B. 61×61
- C. 62×62
- D. 64×64

Correct Answer: C. 62×62

Explanation:

For no padding and stride 1:

$$\begin{aligned} \text{Output Size} &= \text{Input Size} - \text{Filter Size} + 1 \\ \text{Output Size} &= 64 - 3 + 1 = 62 \end{aligned}$$

So, the output feature map size is:

$$62 \times 62$$

4. Applying Max Pooling Output Size

A CNN used for image classification applies max pooling after convolution. The input feature map size is 32×32 . The pooling window size is 2×2 , and stride is 2. The pooling operation reduces the spatial size while preserving important features. Apply the pooling concept and calculate the output size.

- A. 8×8
- B. 16×16
- C. 30×30
- D. 64×64

Correct Answer: B. 16×16

Explanation:

With 2×2 pooling and stride 2:

$$\text{Output Size} = \frac{32}{2} = 16$$

So, the output feature map size is:

$$16 \times 16$$

5. Applying Deep Learning Classification Accuracy

A deep learning model is trained to classify images as cat, dog, or bird. During testing, the model classifies 500 images. Out of these, 450 images are correctly classified. The developer calculates the classification accuracy to evaluate the model performance. Apply the accuracy calculation and determine the classification accuracy.

- A. 80%
- B. 85%
- C. 90%
- D. 95%

Correct Answer: C. 90%

Explanation:

$$\begin{aligned} \text{Accuracy} &= \frac{\text{Correct Predictions}}{\text{Total Images}} \times 100 \\ \text{Accuracy} &= \frac{450}{500} \times 100 = 90\% \end{aligned}$$

So, the classification accuracy is 90%.

1. Applying CNN Convolution Output Size

A deep learning model is used to classify fruit images. The input image size is 64×64 , and the first convolution layer uses a 5×5 filter with stride 1 and no padding. The developer wants to calculate the feature map size after convolution before passing it to the next layer. This helps in designing the CNN architecture correctly. Apply the CNN convolution concept and determine the output feature map size.

- A. 58×58
- B. 59×59
- C. 60×60
- D. 64×64

Correct Answer: C. 60×60

Explanation:

For convolution with no padding:

$$\begin{aligned} \text{Output Size} &= \text{Input Size} - \text{Filter Size} + 1 \\ \text{Output Size} &= 64 - 5 + 1 = 60 \end{aligned}$$

So, the output feature map size is:

$$60 \times 60$$

2. Applying Max Pooling in CNN

A CNN model processes a feature map of size 28×28 after convolution. A max pooling layer with window size 2×2 and stride 2 is applied. The purpose of pooling is to reduce spatial size while retaining important features. The engineer wants to find the reduced feature map size. Use the pooling concept and calculate the output size.

- A. 7×7
- B. 14×14
- C. 26×26
- D. 56×56

Correct Answer: B. 14×14

Explanation:

With 2×2 pooling and stride 2:

$$\text{Output Size} = \frac{28}{2} = 14$$

So, the output feature map size is:

$$14 \times 14$$

3. Applying CNN Classification Accuracy

A CNN is trained to classify medical images as normal or abnormal. During testing, the model checks 800 images. Out of these, 720 images are correctly classified. The hospital wants to measure the model accuracy before using it as a decision-support tool. Apply the accuracy calculation and determine the CNN classification accuracy.

- A. 80%
- B. 85%
- C. 90%
- D. 95%

Correct Answer: C. 90%

Explanation:

$$\begin{aligned} \text{Accuracy} &= \frac{\text{Correct Predictions}}{\text{Total Images}} \times 100 \\ \text{Accuracy} &= \frac{720}{800} \times 100 = 90\% \end{aligned}$$

So, the CNN classification accuracy is 90%.

4. Applying Vision Transformer Patch Count

A Vision Transformer is used for image classification. The input image size is 224×224 , and the image is divided into non-overlapping patches of size 16×16 . Each patch is converted into a token before being processed by the transformer encoder. The developer wants to calculate the total number of image patches. Apply the patch division concept and calculate the number of patches.

- A. 64 patches
- B. 128 patches
- C. 196 patches
- D. 256 patches

Correct Answer: C. 196 patches

Explanation:

Number of patches along width:

$$\frac{224}{16} = 14$$

Number of patches along height:

$$\frac{224}{16} = 14$$

Total patches:

$$14 \times 14 = 196$$

So, the image is divided into 196 patches.

5. Applying Transformer Token Count

A Vision Transformer divides an image into 196 image patches. In addition to these patch tokens, the model adds one special classification token, called the CLS token, for final image classification. The transformer encoder processes all tokens together. Use the Vision Transformer token concept and determine the total number of tokens.

- A. 196 tokens
- B. 197 tokens
- C. 198 tokens
- D. 200 tokens

Correct Answer: B. 197 tokens

Explanation:

$$\begin{aligned} \text{Total Tokens} &= \text{Patch Tokens} + \text{CLS Token} \\ \text{Total Tokens} &= 196 + 1 = 197 \end{aligned}$$

So, the transformer processes 197 tokens.

1. Applying GAN Discriminator Accuracy

A GAN-based image generation system is trained to generate synthetic face images. During one validation round, the discriminator checks 200 images, including real and generated images. It correctly classifies 160 images as either real or fake. The developer wants to calculate the discriminator accuracy to understand how well it distinguishes real images from generated images. Apply the accuracy calculation and determine the discriminator accuracy.

- A. 70%
- B. 75%
- C. 80%
- D. 90%

Correct Answer: C. 80%

Explanation:

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} \times 100$$
$$\text{Accuracy} = \frac{160}{200} \times 100 = 80\%$$

So, the discriminator accuracy is 80%.

2. Applying GAN Generator Improvement

A GAN is used to generate handwritten digit images. In the first training stage, the discriminator correctly identifies 90 out of 100 generated images as fake. After further training, it correctly identifies only 60 out of 100 generated images as fake. The generator is considered improved when it fools the discriminator more often. Apply the GAN concept and calculate how many more generated images fooled the discriminator after training.

- A. 20 images
- B. 30 images
- C. 40 images
- D. 50 images

Correct Answer: B. 30 images

Explanation:

Initially, fake images identified correctly:

$$90$$

So, images that fooled the discriminator initially:

$$100 - 90 = 10$$

After training, fake images identified correctly:

$$60$$

So, images that fooled the discriminator after training:

$$100 - 60 = 40$$

Improvement:

$$40 - 10 = 30$$

So, 30 more generated images fooled the discriminator.

3. Applying Pattern Classification Accuracy

A pattern recognition system classifies leaf images into healthy and diseased categories. During testing, the system checks 500 leaf images. It correctly classifies 425 images. The agriculture team wants to calculate the recognition accuracy before using the system for field monitoring. Apply the classification accuracy concept and determine the accuracy.

- A. 80%
- B. 82%
- C. 85%
- D. 90%

Correct Answer: C. 85%

Explanation:

$$Accuracy = \frac{\text{Correct Classifications}}{\text{Total Samples}} \times 100$$
$$Accuracy = \frac{425}{500} \times 100 = 85\%$$

So, the pattern recognition accuracy is 85%.

4. Applying Mean Feature Value in Statistical Pattern Recognition

A statistical pattern recognition system classifies fruits using size-based features. Four fruit samples have extracted area values of 20, 30, 40, and 50 square units. The system calculates the mean feature value to summarize the class pattern. This mean value can be used for comparison with new fruit samples. Apply the mean calculation and determine the mean feature value.

- A. 30
- B. 35

- C. 40
- D. 45

Correct Answer: B. 35

Explanation:

$$\begin{aligned} \text{Mean} &= \frac{20 + 30 + 40 + 50}{4} \\ \text{Mean} &= \frac{140}{4} = 35 \end{aligned}$$

So, the mean feature value is 35.

. Applying Initial State Probability in HMM

A weather prediction system uses a Hidden Markov Model to identify whether the hidden weather state is Rainy or Sunny. The first observed activity is “carrying umbrella.” The initial probability of Rainy is 0.6, and the probability of observing “carrying umbrella” when it is Rainy is 0.8. The system calculates the first-step probability for the Rainy state. Apply the HMM first-step concept and calculate the probability for the Rainy state.

- A. 0.24
- B. 0.48
- C. 0.60
- D. 0.80

Correct Answer: B. 0.48

Explanation:

$$\begin{aligned} \alpha_1(\text{Rainy}) &= \pi(\text{Rainy}) \times b_{\text{Rainy}}(\text{Umbrella}) \\ \alpha_1(\text{Rainy}) &= 0.6 \times 0.8 = 0.48 \end{aligned}$$

So, the first-step probability for the Rainy state is 0.48.

2. Applying Transition Probability in HMM

A robot navigation system uses an HMM to estimate its hidden location state. If the robot is currently in Room A, the probability of staying in Room A in the next time step is 0.7, and the probability of moving to Room B is 0.3. If the current probability of being in Room A is 0.5, the system calculates the contribution from Room A to Room B in the next step. Use the transition concept and calculate the probability contribution from Room A to Room B.

- A. 0.15
- B. 0.30
- C. 0.35
- D. 0.50

Correct Answer: A. 0.15

Explanation:

$$\begin{aligned}\text{Contribution} &= \text{Current Probability} \times \text{Transition Probability} \\ \text{Contribution} &= 0.5 \times 0.3 = 0.15\end{aligned}$$

So, the contribution from Room A to Room B is 0.15.

3. Applying Observation Probability in Speech Recognition

A speech recognition system uses an HMM to recognize spoken words. One hidden state represents the sound unit “ka.” The probability of observing a particular acoustic feature when the state is “ka” is 0.75. If the probability of being in the “ka” state at that time is 0.4, the system calculates the combined probability for this state and observation. Apply the HMM observation concept and determine the combined probability.

- A. 0.20
- B. 0.25
- C. 0.30
- D. 0.35

Correct Answer: C. 0.30

Explanation:

$$\begin{aligned}\text{Combined Probability} &= \text{State Probability} \times \text{Observation Probability} \\ \text{Combined Probability} &= 0.4 \times 0.75 = 0.30\end{aligned}$$

So, the combined probability is 0.30.

4. Applying Forward Probability Calculation

A gesture recognition system uses an HMM with two hidden states: Gesture Start and Gesture Move. At the previous time step, the probability of being in Gesture Start is 0.4, and the probability of being in Gesture Move is 0.3. The transition probability from Start to Move is 0.5, and from Move to Move is 0.6. The probability of observing the current hand feature in Gesture Move is 0.8. Apply the forward probability concept and calculate the current probability for Gesture Move.

- A. 0.224
- B. 0.304
- C. 0.380
- D. 0.800

Correct Answer: B. 0.304

Explanation:

First calculate total incoming probability to Gesture Move:

$$(0.4 \times 0.5) + (0.3 \times 0.6) \\ = 0.20 + 0.18 = 0.38$$

Then multiply by observation probability:

$$0.38 \times 0.8 = 0.304$$

So, the current probability for Gesture Move is 0.304.

5. Applying Most Likely State Selection

A handwriting recognition system uses an HMM to identify hidden writing states from pen movement observations. At a particular time step, the probabilities of three hidden states are: Stroke Start = 0.25, Stroke Middle = 0.55, and Stroke End = 0.20. The system selects the state with the highest probability as the most likely state. Apply the state selection rule and determine the most likely hidden state.

- A. Stroke Start
- B. Stroke Middle
- C. Stroke End
- D. All states are equally likely

Correct Answer: B. Stroke Middle

Explanation:

The state probabilities are:

$$\begin{aligned} \textit{Stroke Start} &= 0.25 \\ \textit{Stroke Middle} &= 0.55 \\ \textit{Stroke End} &= 0.20 \end{aligned}$$

The highest probability is 0.55, so the most likely hidden state is Stroke Middle.

1. Applying Mixture Weight Probability in GMM

A medical image analysis system uses a Gaussian Mixture Model to classify pixels into different tissue types. For one pixel, the probability from Component 1 is 0.6, and the mixture weight of Component 1 is 0.4. The system calculates the weighted contribution of this component before combining it with other components. Apply the GMM component contribution concept and calculate the weighted probability.

- A. 0.20
- B. 0.24
- C. 0.40
- D. 0.60

Correct Answer: B. 0.24

Explanation:

$$\begin{aligned}\text{Weighted Probability} &= \text{Mixture Weight} \times \text{Component Probability} \\ \text{Weighted Probability} &= 0.4 \times 0.6 = 0.24\end{aligned}$$

So, the weighted probability is 0.24.

2. Applying Total Probability in Gaussian Mixture Model

A fruit sorting system uses a GMM to classify fruits based on color intensity. For one fruit sample, Component 1 gives a weighted probability of 0.18, Component 2 gives 0.32, and Component 3 gives 0.10. The system combines these values to find the total mixture probability for that sample.

Use the GMM mixture concept and determine the total probability.

- A. 0.50
- B. 0.55
- C. 0.60
- D. 0.70

Correct Answer: C. 0.60

Explanation:

$$\begin{aligned}\text{Total Probability} &= 0.18 + 0.32 + 0.10 \\ \text{Total Probability} &= 0.60\end{aligned}$$

So, the total mixture probability is 0.60.

3. Applying Responsibility Calculation in EM

A GMM-based image segmentation system estimates whether a pixel belongs to Region A or Region B. For one pixel, the weighted probability from Region A is 0.30, and the weighted probability from Region B is 0.20. During the E-step of EM, the system calculates the responsibility of Region A for that pixel.

Apply the EM responsibility concept and calculate the responsibility of Region A.

- A. 0.40
- B. 0.50
- C. 0.60
- D. 0.70

Correct Answer: C. 0.60

Explanation:

$$\begin{aligned}\text{Responsibility of A} &= \frac{\text{Weighted Probability of A}}{\text{Total Weighted Probability}} \\ \text{Responsibility of A} &= \frac{0.30}{0.30 + 0.20}\end{aligned}$$

$$\text{Responsibility of A} = \frac{0.30}{0.50} = 0.60$$

So, the responsibility of Region A is 0.60.

4. Applying Mean Update in EM

A GMM is used to cluster pixel intensities in a grayscale image. During the M-step of EM, one Gaussian component is assigned three pixel values with responsibilities already considered, resulting in selected values 40, 50, and 60. The system updates the mean of that Gaussian component using these values.

Apply the EM mean update concept and calculate the updated mean.

- A. 40
- B. 45
- C. 50
- D. 55

Correct Answer: C. 50

Explanation:

$$\begin{aligned} \text{Mean} &= \frac{40 + 50 + 60}{3} \\ \text{Mean} &= \frac{150}{3} = 50 \end{aligned}$$

So, the updated mean is 50.

5. Applying New Mixture Weight in EM

A GMM is applied to segment an image into two regions. In a dataset of 100 pixels, the total responsibility assigned to Component 1 after the E-step is 35. During the M-step, the mixture weight of Component 1 is updated based on its responsibility share in the dataset.

Apply the EM mixture weight update concept and calculate the new mixture weight.

- A. 0.25
- B. 0.35
- C. 0.50
- D. 0.65

Correct Answer: B. 0.35

Explanation:

$$\begin{aligned} \text{New Mixture Weight} &= \frac{\text{Total Responsibility}}{\text{Total Samples}} \\ \text{New Mixture Weight} &= \frac{35}{100} = 0.35 \end{aligned}$$

So, the new mixture weight is 0.35.

1. Applying SVM Decision Function

A computer vision system uses an SVM to classify fruit images as apple or orange. For one test image, the extracted feature value is $x = 4$. The trained SVM has weight $w = 2$ and bias $b = -5$. The system calculates the decision score to classify the image. Apply the SVM decision concept and determine the decision score.

- A. 2
- B. 3
- C. 4
- D. 8

Correct Answer: B. 3

Explanation:

$$\begin{aligned} \text{Decision Score} &= wx + b \\ \text{Decision Score} &= 2(4) - 5 \\ \text{Decision Score} &= 8 - 5 = 3 \end{aligned}$$

So, the decision score is 3.

2. Applying SVM Class Prediction

A medical image classification system uses an SVM to classify tissue images as normal or abnormal. For one image, the SVM decision score is -2.5. The system follows this rule: if the score is positive, classify as abnormal; if the score is negative, classify as normal. Use the given SVM classification rule and select the predicted class.

- A. Abnormal
- B. Normal
- C. Both normal and abnormal
- D. Cannot classify because the score is negative

Correct Answer: B. Normal

Explanation:

Given decision score:

$$-2.5$$

Since the value is negative, the image is classified as Normal according to the given rule.

3. Applying Margin Calculation in SVM

A vision-based defect detection system uses a linear SVM. The separating hyperplane has weight magnitude $\|w\| = 4$. The engineer wants to calculate the margin width to understand how well the classes are separated. A larger margin usually indicates better separation. Apply the SVM margin concept and calculate the margin width.

- A. 0.25
- B. 0.50
- C. 1.00
- D. 2.00

Correct Answer: B. 0.50

Explanation:

For SVM, margin width is:

$$\begin{aligned} \text{Margin} &= \frac{2}{\|w\|} \\ \text{Margin} &= \frac{2}{4} = 0.50 \end{aligned}$$

So, the margin width is 0.50.

4. Applying Graph Edge Weight Calculation

A graph-based image segmentation system represents pixels as graph nodes. Two neighboring pixels have intensity values 70 and 90. The system calculates the absolute intensity difference to assign edge cost between the two pixel nodes. This cost helps decide whether the pixels should belong to the same region or different regions. Apply the graph edge cost concept and calculate the intensity difference.

- A. 10
- B. 20
- C. 30
- D. 40

Correct Answer: B. 20

Explanation:

$$\begin{aligned} \text{Edge Cost} &= |90 - 70| \\ \text{Edge Cost} &= 20 \end{aligned}$$

So, the edge cost is 20.

5. Applying Graph-Based Label Propagation

A graph-based vision system classifies image regions using label propagation. A test node is connected to three labeled neighboring nodes. Two neighbors are labeled Object, and one neighbor is labeled Background. The system assigns the label based on majority voting among connected neighbors. Use the graph-based learning rule and determine the assigned label.

- A. Object
- B. Background
- C. Both labels equally
- D. No label can be assigned

Correct Answer: A. Object

Explanation:

Neighbor labels:

$$\text{Object} = 2$$

$$\text{Background} = 1$$

Since Object has the majority count, the test node is assigned the label Object.

Image based sample question:

1. Analyze Hough Transform Output for Line Detection

A self-driving car uses Hough Transform to detect lane lines from road images. In one frame, the accumulator space shows a strong peak corresponding to a diagonal line. However, the original edge image contains many short broken edge segments due to shadows and road markings. The system still detects a long lane line because many edge points vote for the same line parameter. Analyze the situation and identify which visual option best represents the detected line.

2. Analyze Circle Hough Transform in Industrial Inspection

A bottle cap inspection system uses Circle Hough Transform. The detected object receives high votes for a common center and radius. Another defective cap has an incomplete and irregular boundary, so its votes are scattered in the accumulator space. The system accepts the first cap and rejects the second. Analyze the output and select the shape that would receive the strongest circle votes.

3. Analyze Shape Prior Use in Noisy Segmentation

A medical segmentation system detects kidney boundaries from ultrasound images. In some images, the kidney boundary is weak and partially missing. Without prior information, the segmentation leaks into nearby tissue. When a kidney shape prior is used, the final boundary becomes more realistic and anatomically meaningful. Analyze the result and select the visual option that best represents the effect of using a shape prior.

4. Analyze Shape Correspondence Error

A medical image system compares the shape of an organ in two MRI scans. It places landmark points on both organ boundaries and tries to match corresponding anatomical points. In one result, the landmark near the upper boundary of the first organ is wrongly connected to a lower boundary point of the second organ. This creates a large shape difference even though the organs are actually similar. Analyze the figures and identify the option showing incorrect landmark correspondence.

5. Analyze PCA Direction of Maximum Variation

A face recognition system uses PCA to reduce feature dimensions. The data points are distributed mainly along one dominant direction. PCA selects this direction as the first principal component because it preserves the maximum variation in the data. This helps reduce redundant features while keeping important information. Analyze the figures and identify the option that shows the dominant PCA direction.

6. Analyze CNN and Vision Transformer Processing

A researcher compares CNN and Vision Transformer models for image classification. CNNs focus on local regions using convolution filters, while Vision Transformers divide the image into patches and learn relationships between distant regions through attention. In the experiment, the model is expected to capture global relationships among patches. Analyze the figures and identify the visual representation of Vision Transformer-style patch attention.

7. Analyze GAN Generator Improvement

A GAN is trained to generate synthetic face images. At the beginning, fake images look very different from real ones, so the discriminator easily identifies them. After training, the generated samples become visually closer to real samples, and the discriminator finds it harder to distinguish them. This indicates improvement in the generator. Analyze the figures and identify the option showing improved generator realism.

8. Analyze Graph-Based Learning Structure

A medical image segmentation system represents image regions as nodes. Similar or neighboring regions are connected using edges. The system uses these relationships to assign consistent labels to connected regions. This is useful when nearby or visually similar regions are likely to belong to the same object. Analyze the figures and identify the graph-based learning representation.

9. Analyze SVM Decision Boundary

A vision-based defect classification system extracts two features from product images. The training samples form two groups: normal products and defective products. The model must choose a decision boundary that separates the two groups with a clear margin. A poor boundary may pass close to samples or mix both classes. Analyze the images and select the best SVM decision boundary.

10. Analyze Graph-Based Learning Representation

A vision system segments an image into regions. Each region is treated as a node, and relationships such as similarity, distance, and neighborhood connection are represented as edges. The system uses

these connections to improve region classification. The correct representation should show image elements connected through relationships. Analyze Q2 options in the image sheet and select the graph-based learning representation.

11. Analyze GMM Representation for Overlapping Classes

A medical image contains three types of regions: background, normal tissue, and abnormal tissue. Their pixel intensities overlap, so a single threshold cannot separate them clearly. The system uses a probabilistic model where each tissue group is represented by a smooth distribution, and overlapping regions are handled using soft assignments. The correct visual option should show overlapping statistical distributions rather than isolated points or line voting. Analyze the images sheet and select the option that best represents GMM-based modeling.